

Genetic and environmental contributions to psychopathological symptoms in adulthood: Clarifying the role of individual and parental risk factors

Abstract

Psychopathology-related suicide attempts and deaths are increasing, while the research focus remains on childhood and adolescence. We examined the genetic and environmental contributions to internalizing, externalizing, and general psychopathological distress clarifying the role of individual/parental risk factors in young adults using a publicly-available dataset from the Human Connectome Project, including 1206 young adult. Heritability estimates were analyzed in a subsample of 402 monozygotic and dizygotic twins. The estimated prevalence of psychopathological symptoms ranged from 9%–16%. The heritability of internalizing, externalizing, and general psychopathological distress reached significance with estimates ranging from 22%–46%. Shared (common) environments contributed to aggressive behavior (30%). Household income, childhood conduct problems, and maternal drug or alcohol problems were common risk factors of internalizing, externalizing, and general psychopathological distress. Marijuana dependence and maternal anxiety were additional risk factors of externalizing distress. The presence of alcohol and maternal drug or alcohol problems was linked to general psychopathological distress. The results highlight risk factors associated with psychopathological symptoms and should be considered in early detection of high-risk groups and implementation of family-based interventions. Providing continuous care and/or follow up for high-risk groups from childhood to adulthood may improve mental health and well-being in relatively low- and middle-income groups.

Keywords

Psychopathological symptoms, internalizing, externalizing, distress, twin studies, risk factors, adult, parental, environment

1. Introduction

Mental disorders are broadly prevalent, affecting approximately 17.6–21.1% of the adult population (National Institute on Mental Health, 2017; Thomas et al., 2016). The Global Burden of Disease Study 2015 estimated that in subjects aged 15–39 years, mental disorders were the leading cause of disability, with major depression and anxiety ranked third and the ninth, respectively (GBD 2015 Disease and Injury Incidence and Prevalence Collaborators, 2016). Mental health is regarded as key to human development and also possibly linked to survival with aging, with subjective well-being having a protective role in health (Steptoe et al., 2015; Lyubomirsky et al., 2005). A systematic review revealed that annually, there are 8 million mental disorder-related deaths in the general population, accounting for approximately 14% of all deaths (Walker et al., 2015). In particular, suicidal deaths, one of the most prominent causes of injury mortality (Rockett et al., 2012), comprised 1.49% of global total deaths in 2016 (Naghavi, 2019). Findings from early studies (Arsenault-Lapierre et al., 2004; Nock et al., 2010) indicated a relation between mental illness and suicide, even determining mental disorder as a predictor of suicide attempt. Based on these results and findings of a recent report of a higher mortality rate among individuals with psychiatric disorders (Walker et al., 2015), it is necessary to investigate the risk factors of mental disorders in order to reduce the mortality rate and global burden due to mental-health problems.

The bi-groupings terms of psychopathological problems for clinically referred children (including adolescents), “internalizing” and “externalizing,” were introduced in 1966 and are commonly used in psychiatric diagnoses and psychopathology research (Achenbach, 1966; Achenbach et al., 2016; Cicchetti and Natsuaki, 2014). Although the two terms were initially derived from factor-analytical classification of general psychopathology for children, dimensional scales have been implemented for both children and adults (Achenbach and Edelbrock, 1978; Krueger and Markon, 2006). As prior study has revealed that the internalizing and externalizing dimensions of psychopathology are predictors of suicide attempts in the general population (Hoertel et al., 2015), this study aimed to identify risk factors contributing to these dimensions.

As documented in the Diagnostic and Statistical Manual of Mental Disorders 5 (DSM-5), internalizing problems refer to disorders involving extreme anxiety and depressive, somatic, or obsessive-compulsive symptoms, whereas externalizing problems involve the behavioral patterns of misbehavior or impulsivity without sufficient self-regulation, disruptive conduct, substance use, or attention-deficit/hyperactivity (American Psychiatric Association, 2013). Systematic comorbidities within internalizing/externalizing groups are explained by and considered to be due to both genetic and environmental risk factors (Turker et al., 2019). Using polygenic scores, relevantly recent genome-wide analysis studies have reported on the impact of genetic predisposition on adolescent depressive and conduct disorders (Su et al., 2018) or of pleiotropic genetic variants estimating single-nucleotide polymorphism heritability on general child psychopathology (Neumann et al., 2016). Genetic studies on child or adolescent mental disorders have illustrated the role of genes and/or of the environment on psychopathological phenotypes including internalizing or/and externalizing problems, yet little is known about the specific sub-domains of the psychopathological symptoms. We focused on genetic variability, known as heritability, and on the environmental

variability of subcomponents of internalizing and externalizing symptom, the internalizing and externalizing problems themselves, and general psychopathology using the heritability model to incorporate the dimensional model of psychopathology.

Besides estimating genetic and environmental variation, this study investigated possible risk factors for internalizing and externalizing problems and general psychopathology. We paid particular attention to familial characteristics, such as parental histories of mental problems, relationship status, and individual characteristics, such as alcohol or drug abuse during the lifetime, and conduct problems in childhood. Much of the previous research appears to have focused on the parent-child relationship and on psychiatric disorders in adolescents (Buist et al., 2017; Su et al., 2018). For children and adolescents, sibling relationship quality is related to internalizing and externalizing problems and has a protective role in psychological risk in the familial context (Buist et al., 2013; Keeton et al., 2015). Given that it was shown that childhood behavior problems tended to develop longitudinally into adult psychiatric disorders (Reef et al., 2011) and committed relationships, such as marriage was associated with greater benefits on mental health from a meta-analysis (Braithwaite and Holt-Lunstad, 2017), factors indicating individual's relationship as well as the experience of conduct problems in childhood should be considered among adult population.

Not only impaired parenting (i.e., intrusive or withdrawn parenting) but parental history of depression are determinants of internalizing and externalizing symptoms in children aged 9 to 15 years (Gruhn et al., 2016). Despite some evidence of associations between parental alcohol use or psychiatric disorders and internalizing and externalizing psychopathology in childhood or adulthood (Hser et al., 2015; Miettunen et al., 2014), few studies have focused on an adult sample; thus, there still remains a need to explore evidence pertaining to parental history of neurological or psychiatric disorders contributing to psychopathological symptoms

during adulthood. With respect to individual-level differences along the trajectories of normal and atypical development, it is fundamental to identify processes and mechanisms that lead to the manifestation of psychiatric symptoms and disorder phenotypes (Korous et al., 2018). In a prospective study, cannabis and alcohol use during adolescence preceded internalizing disorders in adulthood after adjusting for earlier emotional and behavioral problems during childhood (Miettunen et al., 2014). A systematic review identified weak average links between internalizing/externalizing behavioral problems and individual socioeconomic status metrics, particularly income and academic background (Korous et al., 2018). This study considered the effects of the current relationship status, sibling structure variables such as monozygotic (MZ¹), dizygotic (DZ²), full siblings of MZ/DZ twins, and full siblings of non-twins; the range of each type of maternal and paternal psychiatric and neurological disorder, and individual facets on internalizing, externalizing, and general psychopathology problems in adulthood.

By using a publicly available dataset from the Human Connectome Project (HCP³) (Glasser et al., 2016), the current study first aimed to assess genetic (A), common or shared (influences that are completely shared between 2 twins, such as living in the same household: C) and non-shared (influences that are unshared between 2 twins in a pair, and unique to each individuals twin, such as having different friends: E) environmental contributions to psychopathological symptoms using a sample of adult twin pairs. The second goal of this study was to verify how strongly individual, parental, and relationship factors are associated with internalizing and externalizing problems of psychopathology and with general mental health problems in adulthood after adjusting for sibling structure effects.

¹ MZ: monozygotic

² DZ: dizygotic

³ HCP: Human Connectome Project

119

120 **2. Materials and methods**

121 **2.1 Participants**

122 The HCP, which agreed to share the data freely, conducted a family-based study of 1206
123 young adults aged 22–35 years from families with twin and non-twin children in conjunction
124 with the WU-Minn Consortium and the McDonnell Center for Systems Neuroscience at
125 Washington University. Approval by the Washington University Institutional review board
126 (IRB) was initially secured for the data collection procedures and sharing of de-identified
127 data (Glasser et al., 2016). The public IRB designated by the Ministry of Health and Welfare
128 of South Korea reviewed our study proposal and determined that it did not require ethical
129 approval (no. P01-201905-21-007).

130 The data of 1090 participants from the HCP 1200 Subjects Release (Glasser et al., 2016) were
131 entered into the analysis after receiving a certificate to use the dataset including Open Access
132 and Restricted Data. Several participants with missing values (110 samples) in the study
133 measures were excluded from the analyses.

134 The demographic characteristics of the 1090 samples are shown in Table 1. The mean age
135 was 28.82 years, and 54% of the participants were female. Approximately 68% of the
136 participants were employed full time, 17.4% were employed part time, and 14.3% were
137 unemployed. The mean annual household income was 50,600 US dollars, and the mean
138 number of years of education was 14.93. Just over half of participants (55.5%) were single or
139 divorced or not cohabiting. Blended or mixed families (a family including children of
140 previous marriages of one spouse or both), at a percentage of 7.2% (78 samples), were
141 excluded from further univariate and multivariate analyses to rule out the influence of

unanticipated events or the environment of remarriage when analyzing heritability estimates. Therefore, 1012 participants (MZ/DZ, full siblings of twins, and full siblings of non-twins) were entered into a univariate analysis and a logistic regression analysis. In terms of the twin genetic modelling analysis, the raw data had to be further refined. After extracting a sample of twins verified by genetic testing and confirming the matched pairs, the data of 402 twins (262 MZs and 140 DZs) were selected to be analyzed as estimates of heritability. The methods used for categorization of the variables are described in the following section.

Table 1. Demographic and socioeconomic characterization of the sample

(1090 participants from the Human Connectome Project 1200 subjects release)		
Characteristic	N (%) or mean $\pm$ SD (range)	
Sex		
Female	591 (54.2)	
Male	499 (45.8)	
Age (years)	28.82 $\pm$ 3.68 (22-37)	
22-25	224 (20.6)	
26-29	384 (35.2)	
≥ 30	482 (44.2)	
Employment Status		
Not working	156 (14.3)	
Part-time employment	190 (17.4)	
Full-time employment	744 (68.3)	
Household income (\$10 thousands)	5.06 $\pm$ 2.15 (1-8)	
< \$50,000	429 (39.4)	
$\geq$ \$50,000	661 (60.6)	
Academic background (years)	14.93 $\pm$ 1.79 (11-17)	
Below upper secondary (≤ 12)	189 (17.3)	
Post-secondary/tertiary (> 12)	901 (82.7)	
Relationship status		
Married or in lived-in relationship	485 (44.5)	
Single or divorced, not cohabitation	605 (55.5)	
Mixed or blended family		
Yes	78 (7.2)	
No	1012 (92.8)	

2.2 Main outcomes variables

2.2.1 Adult psychopathological symptoms

The Achenbach System of Empirically-Based Assessment Adult Self-Report (ASR⁴) (Achenbach, 2009), part of the National Institute of Health Toolbox, was used to assess internalizing and externalizing psychopathological symptoms. The 123-item ASR is a reliable and valid standardized self-report questionnaire intended for completion by adults 18 to 59 years old. It is subdivided into eight scales for behavioral or emotional syndromes: Withdrawal, Somatic Complaints, and Anxiety/Depression, forming an overall Internalizing

⁴ ASR: Achenbach System of Empirically-Based Assessment Adult Self-Report

scale; Rule-Breaking, Aggressive, and Intrusive behavior, forming an overall Externalizing scale; and Thought Problems and Attention Problems. The results of the ASR offer a Total Problem Score (ASR-T⁵) calculated by summing the individual item scores (general psychopathology), the Internalizing score (ASR-I⁶), and the Externalizing score (ASR-E⁷). Higher scores indicate greater distress. In this study, the recommended cutoff scores of the ASR-T, ASR-I, and ASR-E ($60 \leq$, normal; $60-63$, borderline; ≥ 63 , clinical distress) were used to define a psychologically-distressed group with a borderline clinical range compared to a nationally-representative sample (sex and age adjusted T score ≥ 60 , >84 th percentile) (Achenbach, 2009).

2.3 Independent variables

The individual characteristics included were age, sex, household income, employment status, academic background, childhood conduct problems, and psychiatric and substance use history. Participant ages were categorized into two levels: 22 to 29 and 30 to 37 years. Household income was categorized into two sets based on the mean value of the household income (\$50,500): \$10,000 to \$40,000 and \$50,000 to 80,000. Employment status was dichotomized into employed (full time or part time) and unemployed. Academic background was dichotomized into two levels according to the UNESCO international standard classification of education: below upper secondary (≤ 12 years) and post-secondary/tertiary (>12 years). Childhood conduct problems were defined based on a question on the numbers of childhood conduct problems participants reported, with answers categorized into the two

⁵ ASR-T: ASR Total Problem Score

⁶ ASR-I: ASR Internalizing score

⁷ ASR-E: ASR Externalizing score

categories of none and yes (one or more). Psychiatric and substance use history reports (alcohol abuse and alcohol, tobacco, or marijuana dependence) were measured with the DSM-IV criteria of the Semi-Structured Assessment for the Genetics of Alcoholism (Bucholz et al., 1994). Individual histories of alcohol and marijuana problems were defined based on questions whether the participants had met the DSM-IV criteria for alcohol abuse/dependence and marijuana dependence at some point during their lifetimes. Conversely, tobacco dependence was defined based on a question pertaining to the participants having met the DSM-IV criteria for tobacco dependence, including difficulty quitting, tolerance, and withdrawal, at some point during their lifetimes.

A parent with history of certain psychiatric issues (schizophrenia or psychosis, depression, bipolar disorder, anxiety, drug or alcohol problems, Alzheimer's disease or dementia, and Parkinson's disease) was defined if the participants reported that their mother or father had any of the aforementioned disorders. We also considered another factor, i.e., relationship status, that benefits individual psychological well-being by providing instrumental and emotional/social support (Zhu et al., 2018). The relationship status was dichotomized into married or in a live-in relationship and single/divorced or not cohabiting.

Sibling structure variables were derived from the genotyping data (full genome-wide SNP genotyping data) collected, with subsequent confirmation of the mother ID, father ID, and self-reported twin zygosity in the HCP. A twin was defined based on either genotyping data or self-reported zygosity data. The twin zygosity type (i.e., MZ and DZ), as verified by DNA genotyping, was assigned precedence over self-reported zygosity if a conflict existed between self-reported twin zygosity and the genotyping data. After excluding data from participants who were in blended or mixed families, we recategorized a sample of full siblings (full siblings of MZ/DZ twins and full siblings of non-twins). These factors were used to evaluate

heritability estimates in a twin genetic model and were controlled as covariates to investigate the association between individual or parental risk factors and psychopathologic symptoms.

2.4 Statistical analysis

2.4.1 Twin genetic modelling: univariate modelling

First, to build a dataset for the univariate ACE modelling, we used MATLAB (version 2018b, MathWorks, Natick, MA) for data cleaning as follows. From a total of 905 subjects (MZ=272, DZ=152, non-twin=481), after extracting a sample of twins and siblings verified by genetic testing, every subject was assigned a new family ID sequentially incremented starting at number 1 according to the order of the mother ID. The new family IDs were then sorted in order to reveal and remove non-paired twins from the same family. From 272 MZs and 152 DZs, 10 MZs and 12 DZs were excluded, respectively. All 481 non-twin siblings were additionally disregarded in the final dataset and thus the data of the remaining 402 twins (262 MZs and 140 DZs), randomly separated into the dataset, could be analyzed in the ACE model.

Univariate genetic modelling was conducted using OpenMx version 2.12.2 with R version 3.5.3. Univariate modelling was initially used to evaluate the pattern of the relative importance of genetic and environmental effects on variations in the scores of each psychopathology score. We calculated intra-class correlation coefficients (ICCs) with 95% confidence intervals for internalizing (anxiety/depression, withdrawal, and somatic complaint subscales), externalizing (aggressive, rule-breaking, and intrusive behavior subscales) and general psychopathological distress for the 402 twins in this study (131 MZ pairs and 70 DZ pairs), as a subset of the initial HCP dataset. The ICCs were compared to determine an indication of genetic and environmental contributions to the adult psychopathological

symptoms (Plomin et al., 2013). If the correlations are higher for MZ twins than for DZ twins or sibling pairs, a genetic influence (A) is considered to have played a role (Waszczuk et al., 2016). Shared environmental influences (C) are assumed to be evident when the correlations of DZ or sibling pairs are higher than the half of the MZ correlations, and lead to twin similarity over and above genetics (Waszczuk et al., 2016; Plomin et al., 2013). Conversely, a non-shared environment (E) contributes to individual-specific traits that create unique differences among siblings from the same family. This non-shared environmental influence is calculated from the within-pair differences between MZ twins.

We utilized a classical twin model (an ACE model that correspondingly estimates A, C, and E effects) to quantify the genetic and environmental contributions to psychopathological symptoms. In the standard version of the ACE model, the variance of a phenotypic trait can be decomposed into additive genetic (A), shared, or common (C), and environmental and unique (non-shared) environmental effects (E). Heritability, referring to the ratio of the variance of a phenotype attributable to genetic influence, is estimated as twice the difference between monozygotic (MZ) and dizygotic (DZ) twin correlations under the assumption that MZ twin pairs share 100% of their genes, whereas DZ twin pairs and non-twin/full siblings share on average 50% of their segregating genes (Waszczuk et al., 2016; Patel et al., 2017). This model uses the differences in genetic relatedness between MZ twin pairs and DZ twin pairs (Yang et al., 2016). We considered the Akaike information criterion (AIC⁸) and the differences in the -2LL (minus two times log-likelihood) statistics to assess the simplest and most parsimonious model to explain the observed data. The A or C paths were removed, and the best fitting model with the lowest AIC values was subsequently selected (Keyes et al., 2010).

⁸ AIC: Akaike information criterion

2.4.2 Association among demographic characteristics, measured factors, and psychopathological symptoms

We conducted data analysis with SPSS Statistics 25.0 (IBM Co., Armonk, NY) for the descriptive (frequency and standard deviation), chi-square test, and logistic regression analyses. The associations between psychopathological symptoms (internalizing, externalizing, general psychopathology) and their individual, parental, environmental, and genetic correlates were determined in two ways. First, we divided our sample into two groups for each symptom (internalizing, externalizing, and general psychopathological distress/non-distress) and compared these groups using a chi-square test. Differences were considered significant when p values were $<.05$. Second, a logistic analysis was employed to determine the factors contributing to psychopathological symptoms. All independent variables found to be statistically significant with regard to internalizing, externalizing, and general psychopathological distress at the univariate level were entered into a logistic regression model with forward stepwise selection (likelihood-ratio, chi-square model p for entries $=0.05$) (Sokal and Rohlf 1995). The p -values for the logistic regression analyses were obtained after adjustment for the effects of the sibling structures (that is, MZ, DZ, full siblings of MZ/DZ twins, and full siblings of non-twins). Estimated odds ratios and asymptotic 95 percentile intervals are shown. The Hosmer–Lemeshow test was employed to assess the goodness of the fit of the logistic model. All analyses were performed using the 0.05 significance level (two-tailed).

3. Results

3.1 Prevalence of psychopathological symptoms

The ASR results showed that internalizing problems were present in 16%, externalizing problems in 9%, and general health mental problems in 9% of the study participants (Figure 1).

3.2 ICCs for MZ and DZ twins and heritability estimates for psychopathological symptoms

ICCs, heritability (A), and shared (C) and unique environment (E) estimates are presented in Table 2. The genetic effects on internalizing, externalizing, and general psychopathological types of distress were obvious in the ICCs, with the MZ correlations being statistically significantly higher than the DZ correlations. However, the values for the MZ correlations were lower than 1.0, also suggesting environmental effects on the symptom-based phenotypes. Univariate modelling was conducted for internalizing, externalizing, and general psychopathological distress, and in all cases, except for aggressive behavior, the additive genetic-unique environment (AE) model showed the best fit. The heritability estimate (A) was low to moderate (22%–46%) for psychopathologic distress symptoms, while a unique environmental contribution (E) was estimated with a range from 54% to 78% under the AE model. Only the aggressive behavior symptom showed evidence of a shared environment contribution, at 30% in the common and unique environment (CE) model. Fit statistics of univariate models and parameter estimates are presented in the Supplementary table 1.

Table 2. Intraclass correlation coefficients (ICC) for MZ and DZ twins and heritability estimates (N=402)

	ICC MZ (95% CI)	ICC DZ (95% CI)	A (95% CI)	C (95% CI)	E (95% CI)
Internalizing distress	0.43** (0.28, 0.56)	0.15 (-0.08, 0.37)	0.42 (0.28, 0.54)		0.58 (0.46, 0.72)
Anxiety/depressed subscale	0.37** (0.21, 0.51)	0.18 (-0.06, 0.40)	0.36 (0.21, 0.48)		0.64 (0.52, 0.79)
Withdrawn subscale	0.39** (0.24, 0.53)	-0.05 (-0.28, 0.19)	0.33 (0.17, 0.47)		0.67 (0.53, 0.83)
Somatic complaints subscale	0.23* (0.06, 0.39)	0.05 (-0.19, 0.28)	0.22 (0.06, 0.38)		0.78 (0.62, 0.94)
Externalizing distress	0.40** (0.25, 0.54)	-0.01 (-0.24, 0.23)	0.37 (0.22, 0.50)		0.63 (0.50, 0.78)
Aggressive behavior subscale	0.31** (0.15, 0.46)	0.27 (0.03, 0.47)		0.30 (0.16, 0.42)	0.70 (0.58, 0.84)
Rule breaking behavior subscale	0.30** (0.13, 0.45)	-0.02 (-0.26, 0.21)	0.27 (0.10, 0.42)		0.73 (0.58, 0.90)
Intrusive subscale	0.44** (0.29, 0.57)	0.20 (-0.03, 0.42)	0.46 (0.31, 0.58)		0.54 (0.42, 0.69)
General psychopathological distress	0.46** (0.31, 0.58)	0.08 (-0.16, 0.31)	0.43 (0.29, 0.55)		0.57 (0.45, 0.71)

* $p < 0.01$, ** $p < 0.001$. ICC MZ intraclass correlation monozygotic twins, ICC DZ intraclass correlations dizygotic twins. A (heritability), C (shared environment), and E (unique environment).

3.3 Associations between measured factors and psychopathological symptoms

In the univariate analysis, the difference in the frequency of internalizing symptoms was significantly related ($p < .05$) to individual sociodemographic variables (lower household income <\$50,000, childhood conduct problems, satisfaction of the DSM IV alcohol abuse criteria, and of the DSM IV alcohol dependence criteria), and parental factors (a mother or a father having at least more than one instance of history of a psychiatric or neurologic disorders) (Table 3). The difference in the frequency of externalizing symptoms was significantly related to individual sociodemographic variables including the 22–29 age group,

household income lower than \$50,000, an education level lower than or equal to 12 years, childhood conduct problems, and satisfaction of the DSM IV alcohol abuse criteria, DSM IV alcohol dependence criteria, and DSM IV tobacco and DSM IV marijuana dependence criteria. The prevalence of externalizing distress and general psychopathological distress were significantly associated with the parental variables of both maternal and paternal history of psychiatric or neurologic disorders and being married or in a live-in relationship. Moreover, the prevalence of internalizing, externalizing, and general psychopathological distress were significantly associated with the sibling structure variables.

Table 3. Associations between measured factors and psychopathology symptoms (n=1012)

	Internalizing distress n (%)	Non- distress n (%)	Externalizing distress n (%)	Non- distress n (%)	General distress n (%)	Non- distress n (%)
Gender						
Female	78 (49.4)	463 (54.2)	39 (45.3)	502 (54.2)	41 (45.7)	499 (54.2)
Male	80 (50.6)	391 (45.8)	47 (54.7)	424 (45.8)	50 (54.3)	421 (45.8)
<i>p</i> -value		.262		.115		.115
Age (years)						
22-29	94 (59.5)	466 (54.6)	57 (66.3)	503 (54.3)	66 (71.7)	494 (53.7)
≥30	64 (40.5)	388 (45.4)	29 (33.7)	423 (45.7)	26 (28.3)	426 (46.3)
<i>p</i> -value		.252		.033		.001
Household income						
< \$50,000	84 (53.2)	292 (34.2)	52 (60.5)	324 (35.0)	59 (64.1)	317 (34.5)
≥ \$50,000	74 (46.8)	562 (65.8)	34 (39.5)	602 (65.0)	33 (35.9)	603 (65.5)
<i>p</i> -value		<.001		<.001		<.001
Employment						
Not working	25 (19.5)	111 (15.7)	16 (21.9)	120 (15.7)	16 (20.8)	120 (15.8)
Employed	103 (80.5)	598 (84.3)	57 (78.1)	644 (84.3)	61 (79.2)	640 (84.2)
<i>p</i> -value		.274		.169		.258
Academic background						
≤12	25 (15.8)	125 (14.6)	20 (23.3)	130 (14.0)	16 (17.4)	134 (14.6)
> 12	133 (84.2)	729 (85.4)	66 (76.7)	796 (86.0)	76 (82.6)	786 (85.4)
<i>p</i> -value		.715		.021		.467
Married or in lived-in						
No	106 (67.1)	411 (48.1)	58 (67.4)	491 (53.0)	63 (68.5)	486 (52.8)
Yes	52 (32.9)	443 (51.9)	28 (32.6)	435 (47.0)	29 (31.5)	434 (47.2)
<i>p</i> -value		<.001		.010		.004
Childhood conduct prob.						
None	72 (45.6)	548 (64.2)	35 (40.7)	585 (63.2)	39 (42.4)	581 (63.2)
Yes	86 (54.4)	306 (35.8)	51 (59.3)	341 (36.8)	53 (57.6)	339 (36.8)
<i>p</i> -value		<.001		<.001		<.001
Alcohol abuse						
No	124 (78.5)	730 (85.5)	60 (69.8)	794 (85.7)	66 (71.7)	788 (85.7)
Yes	34 (21.5)	124 (14.5)	26 (30.2)	132 (14.3)	26 (28.3)	132 (14.3)
<i>p</i> -value		.026		<.001		<.001
Alcohol dependence						
No	143 (90.5)	810 (94.8)	75 (87.2)	878 (94.8)	82 (89.1)	871 (94.7)
Yes	15 (9.5)	44 (5.2)	11 (12.8)	48 (5.2)	10 (10.9)	49 (5.3)
<i>p</i> -value		.032		.004		.030
Tobacco dependence						
No	119 (75.3)	685 (80.2)	60 (69.8)	744 (80.3)	67 (72.8)	737 (80.1)
Yes	39 (24.7)	169 (19.8)	26 (30.2)	182 (19.7)	25 (27.2)	183 (19.9)
<i>p</i> -value		.162		.020		.099
Marijuana dependence						
No	138 (87.3)	780 (91.3)	66 (76.7)	852 (92.0)	77 (83.7)	841 (91.4)
Yes	20 (12.7)	74 (8.7)	20 (23.3)	74 (8.0)	15 (16.3)	79 (8.6)
<i>p</i> -value		.112		<.001		.015
Family history (mother)						
No	128 (81.0)	747 (87.5)	66 (76.7)	809 (87.4)	74 (80.4)	801 (87.1)
Yes	30 (19.0)	107 (12.5)	20 (23.3)	117 (12.6)	18 (19.6)	119 (12.9)
<i>p</i> -value		.017		.007		.050
Family history (father)						
No	121 (76.6)	707 (82.8)	59 (68.6)	769 (83.0)	66 (71.7)	762 (82.8)
Yes	37 (23.4)	147 (17.2)	27 (31.4)	157 (17.0)	26 (28.3)	158 (17.2)
<i>p</i> -value		.023		.001		.009
Sibling structures						
Monozygotic twin	38 (24.1)	288 (33.7)	15 (17.4)	311 (33.6)	18 (19.6)	308 (33.5)
Dizygotic twin	31 (19.6)	174 (20.4)	14 (16.3)	191 (20.6)	11 (12.0)	194 (21.1)
Full siblings of twins	43 (27.2)	223 (26.1)	27 (31.4)	239 (25.8)	31 (33.7)	235 (25.5)
Full siblings of non-twins	46 (29.1)	169 (19.8)	30 (34.9)	185 (20.0)	32 (34.8)	183 (19.9)
<i>p</i> -value		.024		.001		<.001

General distress = General psychopathological distress, Childhood conduct prob. = Childhood conduct problem, Tobacco dependence = Participant meets/met DSMIV criteria for tobacco dependence (at least one of the following: difficulty quitting, tolerance, withdrawal), Family history (mother) = Mother with at least more than one history of psychiatric or neurologic disorders (schizophrenia or psychosis, depression, bipolar disorder, anxiety, drug or alcohol problems, Alzheimer's disease or dementia, Parkinson's disease), Family history (father) = Father with at least more than one history of psychiatric or neurologic disorders (schizophrenia or psychosis, depression, bipolar disorder, anxiety, drug or alcohol problems, Alzheimer's disease or dementia, Parkinson's disease). *p*-value was calculated by chi-square.

3.4 Selected risk factor models predicting psychopathological symptoms

All independent variables that were found to be statistically significantly related to psychopathologic symptoms (internalizing, externalizing, general psychopathological distress) at the univariate level were entered into the logistic regression model (Table 4). In the binary logistic regression model for predicting internalizing distress (ASR-I scores ≥ 60), household income $< \$50,000$ (OR=2.13, CI: 1.51–3.02), childhood conduct problems (OR=2.09, CI: 1.48–2.96), and maternal drug or alcohol problems (OR=2.71, CI: 1.10–6.67) were all associated with increased risk of problems during adulthood. In the final model for predicting externalizing distress (ASR-E scores ≥ 60), household income $< \$50,000$ (OR=2.30, CI: 1.43–3.70), childhood conduct problems (OR=2.04, CI: 1.27–3.29), satisfaction of the DSM IV marijuana dependence criteria (OR=2.35, CI: 1.30–4.26), maternal history drug or alcohol problems (OR=3.22, CI: 1.11–9.35), and maternal history of anxiety (OR=3.02, CI: 1.24–7.38) were statistically significant. For the sibling structure variables, full siblings of twins (OR=2.12, CI: 1.08–4.17) and full siblings of non-twins (OR=2.68, CI: 1.37–5.27) were associated with an increased risk of internalizing problems. This indicates that the probability of internalizing distress in those who were full siblings of twins was estimated to be greater by 2.12 times, while this was also the case for those who were full siblings of non-twins, where it was estimated to be greater by 2.68 times compared to twins. Seven risk factors were found to be associated with the prevalence of general psychopathological distress (ASR-T ≥ 60): household income $< \$50,000$ (OR=3.23, CI: 2.03–5.15), childhood conduct problems (OR=1.81, CI: 1.14–2.87), alcohol abuse (OR=2.15, CI: 1.25–3.63), alcohol dependence (OR=2.45, CI: 1.12–5.34), maternal drug or alcohol problems (OR=4.14, CI: 1.55–11.03), full siblings of twins (OR=1.92, CI: 1.02–3.60), and full siblings without twin sisters or brothers (OR=2.25, CI: 1.19–4.23).

Table 4. Adjusted odds ratios (ORs) of risk factors for internalizing, externalizing and general psychopathological distress

Risk factor	Internalizing distress	Externalizing distress	General psychopathologic al distress
	Adj. OR (95% CI)		
Individual			
Household income < \$50,000 (ref = ≥ \$50,000)	2.13*** (1.51-3.02)	2.30** (1.43-3.70)	3.23*** (2.03-5.15)
Childhood conduct problems (ref = none)	2.09*** (1.48-2.96)	2.04** (1.27-3.29)	1.81* (1.14-2.87)
Alcohol abuse sometime over lifetime (ref = no)			2.13** (1.25-3.63)
Alcohol dependence sometime over lifetime (ref = no)			2.45* (1.12-5.34)
Marijuana dependence sometime over lifetime (ref = no)		2.35** (1.30-4.26)	
Parental			
Mother with drug/alcohol problems (ref = no)	2.71* (1.10-6.67)	3.22* (1.11-9.35)	4.14** (1.55-11.03)
Mother with anxiety (ref = no)		3.02* (1.24-7.38)	
	N = 1012 (158/854); 84.4% classified correctly, R ² = .041 (Cox & Snell), .070 (Nagelkerke), Model Chi-square = 0.440 (p=.802)	N = 1012 (86/926); 91.5% classified correctly, R ² = .064 (Cox & Snell), .146 (Nagelkerke), Model Chi-square = 9.678 (p=.288)	N = 1012 (92/920); 90.9% classified correctly, R ² = .074 (Cox & Snell), .162 (Nagelkerke); Model Chi-square = 12.398 (p=.134)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (p values were obtained after adjustment for the sibling structure variables such as MZ, DZ, full siblings of MZ/DZ twins, and full siblings of non-twins). CI = confidence interval, OR=odds ratio, internalizing/externalizing/general psychopathological distress were defined as scores of subscale (ASR-I, ASR-E and ASR-T) ≥ 60 . All independent variables found to be statistically significant related to internalizing/externalizing/general psychopathological distress respectively at the univariate level were entered into the logistic regression model with forward stepwise selection routine.

4. Discussion

General psychopathology and internalizing and externalizing disorders are increasingly considered important causes of suicide attempts and deaths worldwide. This study explored the genetic and environmental contributions to internalizing, externalizing, and general psychopathological distress and finally identified individual and parental risk factors associated with psychopathological symptoms in young adults.

First, in the study population, the estimated prevalence of psychopathological symptoms ranged from 9% to 16%, consistent with previous estimates (Fernandez-Pujals et al., 2015). Given the restriction here of the geographic origin of the participants from the HCP dataset with regard to ethnicity/race (more than 90% of the participants were non-Hispanic/Latino in

this study), further studies regarding the prevalence of mental distress in other geographic regions are required.

For the psychopathological types of distress, heritability was found to be highest for general psychopathological distress, estimated at 43%, followed by internalizing distress (42%) and externalizing distress (37%). This estimate differs from those reported by a prior study (Blonigen et al., 2005) of a lower heritability rate for internalizing distress (36%) and a higher rate for externalizing distress (73%) in 17-year-old twins. This discrepancy may be partly explained by the differences in the ages of the participants and the different psychometric instruments used, as internalizing and externalizing behavior problems have been shown to be heritable and genetically stable during childhood and adolescence (Hatoum et al., 2018).

A significant shared environmental influence (30%) without a genetic influence was detected for aggressive behaviors, one of the elements of externalizing distress in this study. Aggressive reactions are normally used to communicate discontent in children and often serve as means of gaining rewards or pleasurable sensations in adolescence (Kuhn et al., 2018). Factors pertaining to aggressive behavior related to common environments shared by siblings or family members (e.g., parenting style and school policy) from adolescence into adulthood should be determined, in that the heritability of impulsivity, associated with externalizing behavior problems, slightly decreases over adolescence, with environmental influences becoming more powerful (Kuhn et al., 2018). In contrast, the genetic influence of intrusive behavior, another element of externalizing distress in this study, was significant (46%). Therefore, it is important to replicate these findings using a longitudinal design but with the same instruments to elucidate how genetic and environmental effects on psychopathological symptoms can change developmental trajectories.

A household income lower than \$50,000, history of childhood conduct problems at the individual level, and maternal drug or alcohol problems were common factors significantly increasing the risk of internalizing, externalizing, and general psychopathological distress in young adults. Previous experience of conduct problems and maternal drug or alcohol problems appeared to relatively strongly and persistently affect all types of psychopathological distress from childhood into adulthood in the current study. It is known that conduct disorder, characterized by aggression to others, the destruction of property, deceitfulness, theft, and rule-breaking behavior during childhood and adolescence, is affected by a complex matrix of genetic and environmental factors (Salvatore and Dick, 2018). In this light, our results suggest that efforts to investigate familial, contextual factors aggravating and buffering child conduct problems are needed. Moreover, mental health professionals should consider both socioeconomic status and childhood conduct problems as risk factors when diagnosing internalizing, externalizing, and general mental health problems during a consultation. An early family-based intervention consisting of coping strategies and resilience and the management of negative feelings would appear to be beneficial to patients ranging from children to young adults.

Among the parental histories of psychiatric or neurologic disorders (schizophrenia or psychosis, depression, bipolar disorder, anxiety, drug or alcohol problems, Alzheimer's disease or dementia, or Parkinson's disease), only maternal substance use (drug/alcohol) and anxiety problems were significant predictors of psychopathological symptoms in adulthood. In terms of externalizing distress, we additionally identified that individual marijuana dependence and maternal anxiety disorder increased the likelihood of misbehavior problems in adulthood. Our findings contradict those reported by a longitudinal study from California, where maternal substance abuse problems with comorbid mental disorders (i.e., depression, anxiety, psychosis, and bipolar disorder) were a significant predictor of child internalizing

behaviors, but not of externalizing behaviors (Hser et al., 2015). However, our findings are consistent with those of a prior study suggesting that regular cannabis users in both adolescence and young adulthood showed elevated externalizing disorders (Foster et al., 2018). Adults with maternal drug or alcohol problems showed a significantly increased risk of externalizing distress by three times or higher, rendering this factor the most powerful predictor of externalizing distress in our model. This finding is in line with that of a prior study (Gerra et al., 2019) suggesting that cannabis use susceptibility is associated with both genetic and environmental risk factors, implying that a primary role of environmental factors, including parental care, should be highlighted. Our findings are also corroborated by a prior study from Israel, which indicated that maternal anxiety symptoms directly predicted offspring externalizing behavior and that marital dissatisfaction in both the mother and father mediated this relationship (Gamliel et al., 2018). In addition, the types of risk/protective factors operating as common environmental factors for the components of externalizing distress (e.g., aggressive behavior) must be discussed in further studies. Not only maternal/individual mental health but family functioning or relationships should be considered when treating adults with externalizing distress.

Our logistic regression model revealed five risk factors for general psychopathology, as calculated from the ASR-T by summing the eight scales of behavioral or emotional syndromes (withdrawal, somatic complaints, anxiety/depression, rule-breaking, aggressive, and intrusive behavior; and thought and attention problems). These were a household income lower than \$50,000, childhood conduct problems, alcohol abuse, alcohol dependence, and maternal drug/alcohol problems. Subjects who had maternal drug or alcohol problems were at a greater risk by about 4.14 times to develop general psychopathological distress than were subjects who did not have such maternal issues. This result is supported by the findings of a Danish cohort study that suggested that maternal alcohol abuse disorder was associated with

a greater risk of offspring alcohol use disorder irrespective of other parental mental disorders (Holst et al., 2019). As the concurrent use of tobacco, cannabis, cocaine, and other illicit drugs among those with alcohol dependence was related to greater severity of psychopathology (Moss et al., 2015), individual and maternal specific patterns of substance use should be assessed as risk factors of psychopathological symptoms in further studies and in clinical practice. Finally, with regard to socioeconomic status, household incomes in young adulthood need to be considered as the most important screening indicator exacerbating psychopathological symptoms.

The present study has some advantages and limitations. Most previous estimates of the heritability of psychopathological symptoms have largely been based on child or adolescent samples (Blonigen et al., 2005; Kuhn et al., 2018; Hatoum et al., 2018), although childhood behavioral problems may longitudinally develop into psychiatric disorders during adulthood (Reef et al., 2011). Our results are informative as they provide valid evidence of the genetic and environmental effects on internalizing, externalizing, and general psychopathologic types of distress, including their specific symptoms, i.e., anxiety, withdrawal, somatic complaints, aggressiveness, rule-breaking, and intrusive behaviors, with a large adult sample. The twin and sibling samples were homogenous in terms of ethnicity as well as representative of original nuclear families (controlling for confounding effects by excluding cases of blended or mixed families) and were well classified by twin zygosity as confirmed by genetic testing. However, efforts to identify gene–environment interplay on psychopathological symptoms must be highlighted in future studies owing to the limitations of the classical twin design, which may slightly overestimate heritability estimates by disregarding gene–environment interactions.

Furthermore, our study expands our understanding of the associations among individual and

parental factors that increase the risk of adult psychopathological symptoms following an adjustment of differences in the sibling structure variables (i.e., twin zygosity/siblings of twins and non-twins). However, given that we studied individuals from the general population who were evaluated and voluntarily reported their psychopathological symptoms to validate these associations, we acknowledge that these findings may not generalize to a clinically-diagnosed sample. In the present study, to focus on the presence or absence of risk factors associated with psychopathological symptoms, the number of childhood conduct problems reported, the DSM criteria of the individual histories of alcohol, tobacco, marijuana problems; and numerous parental histories of psychiatric or neurologic disorders were coded into categorical, not continuous, variables, implying that there could be ambiguity with regard to the severity or duration of the risk factors. Future studies with longitudinal twin designs, objective measures, and/or larger or more diverse clinically-diagnosed samples are required to address these issues.

5. Conclusions

In summary, psychopathological symptoms such as those associated with internalizing, externalizing, and general psychopathological types of distress in adulthood are substantially heritable and show similar risk associations with household income, childhood conduct problems, and maternal drug/alcohol problems. Externalizing distress is also linked to individual marijuana dependence and maternal anxiety problems, while a shared family environment was found to be a significant component of aggressive behavior. In terms of general psychopathological distress, individual and maternal specific patterns of the concurrent use of substances, particularly alcohol, and other illicit drugs should be assessed and treated by means of early intervention. Although internalizing, externalizing, and overall

behavioral problems in children and adolescents may continue to develop in adulthood, these results are suggestive of such an association and show that the non-shared (unique) environmental influence of parenting does in most cases have greater impact on adult psychopathological symptoms compared to genetics or a shared environment. The implementation of effective family-based interventions targeting these risk factors and the provision of continuing care or follow up for high-risk groups during childhood and adolescence have the potential to improve mental health and well-being in relatively low- and middle-income groups.

Authors disclosures: none o

Declaration of Competing Interest: This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Acknowledgements

The data were provided by the Human Connectome Project (HCP), WU-Minn Consortium and by the McDonnell Center for Systems Neuroscience at Washington University. We thank the participants and their families who voluntarily engaged in the HCP.

References

- Achenbach, T.M., 1966. The classification of children's psychiatric symptoms: a factor-analytic study. *Psychol Monogr.* 80, 1-37. <https://doi.org/10.1037/h0093906>
- Achenbach, T.M., Edelbrock, C.S., 1978. The classification of child psychopathology: A review and analysis of empirical efforts. *Psychol Bull.* 85, 1275-1301. <https://doi.org/10.1037/0033-2909.85.6.1275>

545 Achenbach, T.M., Ivanova, M.Y., Rescorla, L.A., Turner, L.V., Althoff, R.R., 2016.
 546 Internalizing/Externalizing Problems: Review and Recommendations for Clinical and
 547 Research Applications. *J. Am Acad Child Psy.* 55, 647–656.
 548 <https://doi.org/10.1016/j.jaac.2016.05.012>

549 American Psychiatric Association. 2013. Diagnostic and statistical manual of mental
 550 disorders DSM-5. American Psychiatric Pub, Arlington, VA.

551 Arseneault-Lapierre, G., Kim, C., Turecki, G., 2004. Psychiatric diagnoses in 3275 suicides: A
 552 meta-analysis. *BMC Psychiatry.* 4, 37. <https://doi.org/10.1186/1471-244X-4-37>

553 Blonigen, D.M., Hicks, B.M., Krueger, R.F., Patrick, C.J., Iacono, W.G., 2005. Psychopathic
 554 personality traits: Heritability and genetic overlap with internalizing and externalizing
 555 psychopathology. *Psychol Med.* 35, 637-48. <https://doi.org/10.1017/S0033291704004180>

556 Braithwaite, S., Holt-Lunstad, J., 2017. Romantic relationships and mental health. *Current*
 557 *Opinion in Psychology.* 13, 120-125. <https://doi.org/10.1016/j.copsyc.2016.04.001>

558 Bucholz, K.K., Cadoret, R., Cloninger, C.R., Dinwiddie, S.H., Hesselbrock, V.M.,
 559 Nurnberger, J.I., Reich, T., Schmidt, I., Schuckit, M.A., 1994. A new, semi-structured
 560 psychiatric interview for use in genetic linkage studies: A report on the reliability of the
 561 SSAGA. *J. Stud. Alcohol.* 55, 149-58. <https://doi.org/10.15288/jsa.1994.55.149>

562 Buist, K.L., Verhoeven, M., Hoksbergen, R., ter Laak, J., Watve, S., Paranjpe, A., 2017.
 563 Associations of Perceived Sibling and Parent-Child Relationship Quality With Internalizing
 564 and Externalizing Problems: Comparing Indian and Dutch Early Adolescents. *J. Early*
 565 *Adolesc.* 37, 1163-1190. <https://doi.org/10.1177/0272431616653473>

566 Cicchetti, D., Natsuaki, M.N., 2014. Multilevel developmental perspectives toward
 567 understanding internalizing psychopathology: Current research and future directions. *Dev.*

568 Psychopathol. 26, 1189-1190. <https://doi.org/10.1017/S0954579414000959>

569 Fernandez-Pujals, A.M., Adams, M.J., Thomson, P., McKechnie, A.G., Blackwood, D.H.R.,
570 Smith, B.H., Dominiczak, A.F., Morris, A.D., Matthews, K., Campbell, A., Linksted, P.,
571 Haley, C.S., Deary, I.J., Porteous, D.J., MacIntyre, D.J., McIntosh, A.M., 2015.
572 Epidemiology and heritability of major depressive disorder, stratified by age of onset, sex,
573 and illness course in generation Scotland: Scottish family health study (GS: SFHS). PLoS
574 One. 10, e0142197. <https://doi.org/10.1371/journal.pone.0142197>

575 Foster, K.T., Arterberry, B.J., Iacono, W.G., McGue, M., Hicks, B.M., 2018. Psychosocial
576 functioning among regular cannabis users with and without cannabis use disorder. Psychol.
577 Med. 48, 1853-1861. <https://doi.org/10.1017/S0033291717003361>

578 Gamliel, K.H., Dollberg, D.G., Levy, S., 2018. Relations Between Parents' Anxiety
579 Symptoms, Marital Quality, and Preschoolers' Externalizing and Internalizing Behaviors. J.
580 Child. Fam. Stud. 27, 3952-3963. <https://doi.org/10.1007/s10826-018-1212-3>

581 GBD 2015 Disease and Injury Incidence and Prevalence Collaborators, 2016. Global,
582 regional, and national incidence, prevalence, and years lived with disability for 310 diseases
583 and injuries, 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015.
584 Lancet. 388, 1545-1602. [https://doi.org/10.1016/S0140-6736\(16\)31678-6](https://doi.org/10.1016/S0140-6736(16)31678-6)

585 Gerra, M.C., Manfredini, M., Cortese, E., Antonioni, M.C., Leonardi, C., Magnelli, F.,
586 Somaini, L., Jayanthi, S., Cadet, J.L., Donnini, C., 2019. Genetic and Environmental Risk
587 Factors for Cannabis Use: Preliminary Results for the Role of Parental Care Perception. Subst.
588 Use. Misuse. 54, 670-680. <https://doi.org/10.1080/10826084.2018.1531430>

589 Glasser, M.F., Smith, S.M., Marcus, D.S., Andersson, J.L.R., Auerbach, E.J., Behrens, T.E.J.,
590 Coalson, T.S., Harms, M.P., Jenkinson, M., Moeller, S., Robinson, E.C., Sotiropoulos, S.N.,

591 Xu, J., Yacoub, E., Ugurbil, K., Van Essen, D.C., 2016. The Human Connectome Project's
 592 neuroimaging approach. *Nat. Neurosci.* 19, 1175-1187. <https://doi.org/10.1038/nn.4361>

593 Gruhn, M.A., Dunbar, J.P., Watson, K.H., Reising, M.M., McKee, L., Forehand, R., Cole,
 594 D.A., Compas, B.E., 2016. Testing specificity among parents' depressive symptoms,
 595 parenting, and child internalizing and externalizing symptoms. *J. Fam. Psychol.* 30, 309-319.
 596 <https://doi.org/10.1037/fam0000183>

597 Hatoum, A.S., Rhee, S.H., Corley, R.P., Hewitt, J.K., Friedman, N.P., 2018. Etiology of
 598 Stability and Growth of Internalizing and Externalizing Behavior Problems Across Childhood
 599 and Adolescence. *Behav Genet.* 48, 298-314. <https://doi.org/10.1007/s10519-018-9900-8>

600 Hoertel, N., Franco, S., Wall, M.M., Oquendo, M.A., Kerridge, B.T., Limosin, F., Blanco, C.,
 601 2015. Mental disorders and risk of suicide attempt: A national prospective study. *Mol.*
 602 *Psychiatry.* 20, 718-726. <https://doi.org/10.1038/mp.2015.19>

603 Holst, C., Tolstrup, J.S., Sorensen, H.J., Pisinger, V.S.C., Becker, U., 2019. Parental alcohol
 604 use disorder with and without other mental disorders and offspring alcohol use disorder. *Acta.*
 605 *Psychiatr. Scand.* 139, 508-517. <https://doi.org/10.1111/acps.13002>

606 Hser, Y.I., Lanza, H.I., Li, L., Kahn, E., Evans, E., Schulte, M., 2015. Maternal Mental
 607 Health and Children's Internalizing and Externalizing Behaviors: Beyond Maternal
 608 Substance Use Disorders. *J. Child. Fam. Stud.* 24, 638-648. [https://doi.org/10.1007/s10826-](https://doi.org/10.1007/s10826-013-9874-3)
 609 [013-9874-3](https://doi.org/10.1007/s10826-013-9874-3)

610 Keeton, C.P., Teetsel, R.N., Dull, N.M.S., Ginsburg, G.S., 2015. Parent Psychopathology and
 611 Children's Psychological Health: Moderation by Sibling Relationship Dimensions. *J.*
 612 *Abnorm. Child. Psychol.* 43, 1333-1342. <https://doi.org/10.1007/s10802-015-0013-z>

613 Keyes, C.L.M., Myers, J.M., Kendler, K.S., 2010. The structure of the genetic and

614 environmental influences on mental well-being. *Am. J. Public. Health.* 100, 2379-2384.
615 <https://doi.org/10.2105/AJPH.2010.193615>

616 Korous, K.M., Causadias, J.M., Bradley, R.H., Luthar, S.S., 2018. Unpacking the link
617 between socioeconomic status and behavior problems: A second-order meta-analysis. *Dev.*
618 *Psychopathol.* 30, 1889-1906. <https://doi.org/10.1017/S0954579418001141>

619 Krueger, R.F., Markon, K.E., 2006. Reinterpreting Comorbidity: A Model-Based Approach
620 to Understanding and Classifying Psychopathology. *Annu. Rev. Clin. Psychol.* 2, 111–133.
621 <https://doi.org/10.1146/annurev.clinpsy.2.022305.095213>

622 Kuhn, M.A., Ahles, J.J., Aldrich, J.T., Wielgus, M.D., Mezulis, A.H., 2018. Physiological
623 Self-Regulation Buffers the Relationship between Impulsivity and Externalizing Behaviors
624 among Nonclinical Adolescents. *J. Youth. Adolesc.* 47, 829-841.
625 <https://doi.org/10.1007/s10964-017-0689-1>

626 Lyubomirsky, S., King, L., Diener, E., 2005. The benefits of frequent positive affect: Does
627 happiness lead to success? *Psychol. Bull.* 131, 803-855. [https://doi.org/10.1037/0033-](https://doi.org/10.1037/0033-2909.131.6.803)
628 [2909.131.6.803](https://doi.org/10.1037/0033-2909.131.6.803)

629 Miettunen, J., Murray, G.K., Jones, P.B., Mäki, P., Ebeling, H., Taanila, A., Joukamaa, M.,
630 Savolainen, J., Törmänen, S., Järvelin, M.R., Veijola, J., Moilanen, I., 2014. Longitudinal
631 associations between childhood and adulthood externalizing and internalizing
632 psychopathology and adolescent substance use. *Psychol. Med.* 44, 1727-1738.
633 <https://doi.org/10.1017/S0033291713002328>

634 Moss, H.B., Goldstein, R.B., Chen, C.M., Yi, H.Y., 2015. Patterns of use of other drugs
635 among those with alcohol dependence: Associations with drinking behavior and
636 psychopathology. *Addict. Behav.* 50, 192-198. <https://doi.org/10.1016/j.addbeh.2015.06.041>

637 Naghavi M., 2019. Global, regional, and national burden of suicide mortality 1990 to 2016:
638 systematic analysis for the Global Burden of Disease Study 2016. *BMJ*. 364, 194.

639 National Institute on Mental Health, 2017. Mental illness prevalence from the 2017 National
640 Survey on Drug Use and Health. [http://www.nimh.nih.gov/health/statistics/prevalence/any-](http://www.nimh.nih.gov/health/statistics/prevalence/any-mental-illness-ami-among-adults.shtml)
641 [mental-illness-ami-among-adults.shtml](http://www.nimh.nih.gov/health/statistics/prevalence/any-mental-illness-ami-among-adults.shtml) (accessed 15 April 2019).

642 Neumann, A., Pappa, I., Lahey, B.B., Verhulst, F.C., Medina-Gomez, C., Jaddoe, V.W.,
643 Bakermans-Kranenburg, M.J., Moffitt, T.E., van IJzendoorn, M.H., Tiemeier, H., 2016.
644 Single Nucleotide Polymorphism Heritability of a General Psychopathology Factor in
645 Children. *J. Am. Acad. Child. Adolesc. Psychiatry*. 55, 1038-1045.
646 <https://doi.org/10.1016/j.jaac.2016.09.498>

647 Nock, M.K., Hwang, I., Sampson, N.A., Kessler, R.C., 2010. Mental disorders, comorbidity
648 and suicidal behavior: Results from the national comorbidity survey replication. *Mol*
649 *Psychiatry*. 15, 868-876. <https://doi.org/10.1038/mp.2009.29>

650 Patel, S., Park, M.T.M., Devenyi, G.A., Patel, R., Masellis, M., Knight, J., Chakravarty,
651 M.M., 2017. Heritability of hippocampal subfield volumes using a twin and non-twin siblings
652 design. *Hum. Brain. Mapp*. 38, 4337-4352. <https://doi.org/10.1002/hbm.23654>

653 Plomin, R., DeFries, J.C., Knopik, V.S., Neiderhiser, J.M., 2013. Behavioral Genetics, 6th ed.
654 Worth Publisher, New York. <https://doi.org/10.1007/s10519-013-9598-6>

655 Reef, J., Diamantopoulou, S., Van Meurs, I., Verhulst, F.C., Van Der Ende, J., 2011.
656 Developmental trajectories of child to adolescent externalizing behavior and adult DSM-IV
657 disorder: Results of a 24-year longitudinal study. *Soc. Psychiatry. Psychiatr. Epidemiol*. 46,
658 1233-1241. <https://doi.org/10.1007/s00127-010-0297-9>

659 Rockett, I.R.H., Regier, M.D., Kapusta, N.D., Coben, J.H., Miller, T.R., Hanzlick, R.L., Todd,

660 K.H., Sattin, R.W., Kennedy, L.W., Kleinig, J., Smith, G.S., 2012. Leading causes of
 661 unintentional and intentional injury mortality: United States, 2000-2009. *Am. J. Public.*
 662 *Health.* 102, e84-92. <https://doi.org/10.2105/AJPH.2012.300960>

663 Salvatore, J.E., Dick, D.M., 2018. Genetic influences on conduct disorder. *Neurosci.*
 664 *Biobehav. Rev.* 91, 91-101. <https://doi.org/10.1016/j.neubiorev.2016.06.034>

665 Sokal, R.R., Rohlf, F.J., 1995. *Biometry: The principles and practice of statistics in biological*
 666 *research*, 3rd ed. W. H. Freeman, New York.

667 Steptoe, A., Deaton, A., Stone, A.A., 2015. Subjective wellbeing, health, and ageing. *Lancet.*
 668 385, 640-648. [https://doi.org/10.1016/S0140-6736\(13\)61489-0](https://doi.org/10.1016/S0140-6736(13)61489-0)

669 Su, J., Kuo, S.I.C., Bucholz, K.K., Edenberg, H.J., Kramer, J.R., Schuckit, M., Dick, D.M.,
 670 2018. Understanding Mechanisms of Genetic Risk for Adolescent Internalizing and
 671 Externalizing Problems: The Mediating Role of Parenting and Personality. *Twin. Res. Hum.*
 672 *Genet.* 21, 310–321. <https://doi.org/10.1017/thg.2018.36>

673 Thomas, S., Jenkins, R., Burch, T., Nasir, L.C., Fisher, B., Giotaki, G., Gnani, S., Hertel, L.,
 674 Marks, M., Mathers, N., Millington-Sanders, C., Morris, D., Ruprah-Shah, B., Stange, K.,
 675 Thomas, P., White, R., Wright, F., 2016. Promoting mental health and preventing mental
 676 illness in general practice. *London. J. Prim. Care (Abingdon).* 8, 3-9.
 677 <https://doi.org/10.1080/17571472.2015.1135659>

678 Tucker, D.M., Poulsen, C., Luu, P., 2015. Critical periods for the neurodevelopmental
 679 processes of externalizing and internalizing. *Dev. Psychopathol.* 27, 321–346.
 680 <https://doi.org/10.1017/S0954579415000024>

681 Walker, E.R., McGee, R.E., Druss, B.G., 2015. Mortality in mental disorders and global
 682 disease burden implications a systematic review and meta-analysis. *JAMA Psychiatry.* 72,

334–341. <https://doi.org/10.1001/jamapsychiatry.2014.2502>

Waszczuk, M.A., Leonard, H.C., Hill, E.L., Rowe, R., Gregory, A.M., 2016. Coordination difficulty and internalizing symptoms in adults: A twin/sibling study. *Psychiatry. Res.* 239, 1-8. <https://doi.org/10.1016/j.psychres.2016.02.044>

Yang, Z., Zuo, X.N., McMahon, K.L., Craddock, R.C., Kelly, C., De Zubicaray, G.I., Hickie, I., Bandettini, P.A., Castellanos, F.X., Milham, M.P., Wright, M.J., 2016. Genetic and Environmental Contributions to Functional Connectivity Architecture of the Human Brain. *Cereb. Cortex.* 26, 2341-2352. <https://doi.org/10.1093/cercor/bhw027>

Zhu, X., Wang, K., Chen, L., Cao, A., Chen, Q., Li, J., Qiu, J., 2018. Together Means More Happiness: Relationship Status Moderates the Association between Brain Structure and Life Satisfaction. *Neuroscience.* 384, 406-416. <https://doi.org/10.1016/j.neuroscience.2018.05.018>

Figure legends

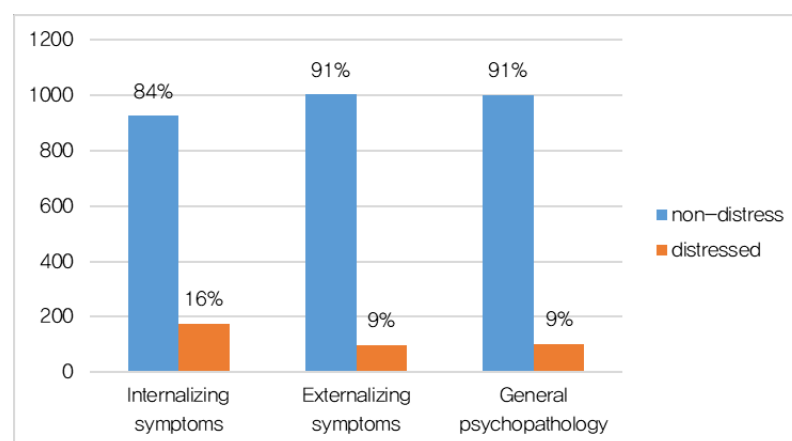


Figure 1. Prevalence of psychopathology symptoms

